

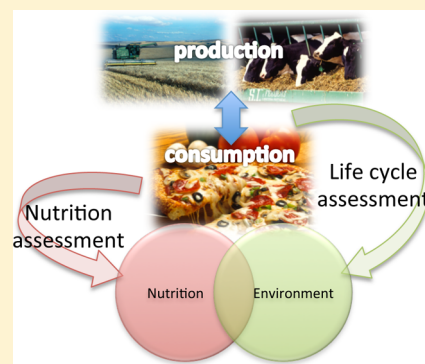
Toward a Life Cycle-Based, Diet-level Framework for Food Environmental Impact and Nutritional Quality Assessment: A Critical Review

Martin C. Heller,^{*,†} Gregory A. Keoleian,[†] and Walter C. Willett[‡]

[†]Center for Sustainable Systems, School of Natural Resources and Environment, University of Michigan, 3012 Dana Building, 440 Church Street, Ann Arbor, Michigan 48109-1041, United States

[‡]Departments of Nutrition and Epidemiology, Harvard School of Public Health, Boston, Massachusetts 02138, United States

ABSTRACT: Supplying adequate human nutrition within ecosystem carrying capacities is a key element in the global environmental sustainability challenge. Life cycle assessment (LCA) has been used effectively to evaluate the environmental impacts of food production value chains and to identify opportunities for targeted improvement strategies. Dietary choices and resulting consumption patterns are the drivers of production, however, and a consumption-oriented life cycle perspective is useful in understanding the environmental implications of diet choices. This review identifies 32 studies that use an LCA framework to evaluate the environmental impact of diets or meals. It highlights the state of the art, emerging methodological trends and current challenges and limitations to such diet-level LCA studies. A wide range of bases for analysis and comparison (i.e., functional units) have been employed in LCAs of foods and diet; we conceptually map appropriate functional unit choices to research aims and scope and argue for a need to move in the direction of a more sophisticated and comprehensive nutritional basis in order to link nutritional health and environmental objectives. Nutritional quality indices are reviewed as potential approaches, but refinement through ongoing collaborative research between environmental and nutritional sciences is necessary. Additional research needs include development of regionally specific life cycle inventory databases for food and agriculture and expansion of the scope of assessments beyond the current focus on greenhouse gas emissions.



1. INTRODUCTION

Nutrition is a fundamental human need, and access to sufficient and proper nutrition affects health and well-being throughout the lifespan in a myriad of ways. A vast and interconnected array of physical, social, and political systems, known collectively as the “food system,” assembles to supply nutrition, and increasingly it seems, permit the paradoxical coexistence of malnutrition and obesity, often within the same population.¹ Concurrently, supplying nutrition in its current form to seven billion humans may be breaching the finite capacity of our planet; a broad scientific agreement has emerged that the food system both illustrates and is a key element in the challenge of global environmental sustainability.^{2,3} In developed countries, food consumption contributes between 15% and 28% to overall national greenhouse gas emissions (GHGE).⁴ Agriculture is responsible for 70–80% of all human water withdrawals and contributes significantly to water pollution.⁵ Agricultural expansion, particularly in the tropics, is a dominant cause of biodiversity loss.³ Intensified use of fertilizers have dramatically disrupted global nitrogen and phosphorus cycles, impacting water quality, aquatic ecosystems, and marine fisheries.^{6,7} The daunting challenge at hand, therefore, is to refashion the food system to deliver better nutritional outcomes to a rapidly growing global population at reduced environmental cost.

Perspectives on how this can be achieved are diverse and often divergent. In 2003, Heller and Keoleian⁸ offered a life-cycle based approach to assessing sustainability of the U.S. food system; they emphasized the importance of reconnecting consumption behaviors with production practices and highlighted the impacts of overconsumption and food wastage. Garnett⁹ recently identified three emerging tendencies, or perspectives, on approaching food sustainability, defining them as efficiency oriented, demand restraint, and food system transformation. The efficiency-oriented perspective focuses on food production and food producers and envisions technological innovations and managerial changes as key to achieving food system sustainability. Demand constraint, on the other hand, sees the problem lying with consumers and unsustainable consumption patterns, calling for reduced consumption of high impact foods. The third perspective, food system transformation, acknowledges the need for socio-economic structural change to achieve social justice and environmental sustainability. Garnett acknowledges that each perspective has its strengths and weaknesses, and that a composite approach

Received: June 5, 2013

Revised: October 2, 2013

Accepted: October 23, 2013

Published: October 23, 2013

that draws on all three perspectives will surely be needed to address mounting challenges.

Common to any perspective for moving toward a more sustainable food system is a need to evaluate indicators of progress within a generally agreed upon methodological framework. Recent trends in nutrition science and nutritional epidemiology have been away from a focus on single dietary components and toward an evaluation of whole diets and dietary patterns as indicators of healthy nutrition.^{10,11} Similarly in recent decades, environmental science and evaluation of the environmental impact of consumer products have moved toward more holistic, comprehensive approaches, and life cycle assessment (LCA) has emerged as a dominant methodological framework.¹² This review primarily addresses the methodological shifts in life cycle assessment approaches that have evolved, and in our opinion, should be enhanced and encouraged, when adopting a consumption-oriented (demand restraint) perspective in working toward a sustainable food system. While a host of legitimate methodological frameworks have emerged, we focus here on literature based in an LCA approach because of its comprehensive perspective.

2. LIFE CYCLE ASSESSMENT OF AGRICULTURAL PRODUCTS AND FOOD ITEMS

LCA is a tool to assess the potential environmental impacts of product systems and services, accounting for the emissions and resource use throughout a product's life cycle, that is, from raw material acquisition through production, distribution, use, and disposal.¹³ While LCA has been defined and standardized through international guidelines,^{13,14} there remains great flexibility in the method, thus permitting application to a wide range of questions about diverse product and service systems. The basic LCA framework is an iterative procedure involving: definition of the goal and scope of the study—what are we studying, how are we studying it, why, and for whom?; life cycle inventory analysis—data collection and calculation procedures to quantify relevant inputs and outputs (energy, raw materials, coproducts, waste, emissions to air, water, and soil) across each unit process within the system boundary; life cycle impact assessment—associating inventory data with specific environmental impact categories and modeling the relevance of those impacts; and interpretation of outcomes.

Agricultural and food product systems have offered both an ideal and challenging application of LCA methods due to their complexity and their close interlink between nature and the technical sphere. Figure 1 gives an indication of the growth of food-related LCA articles in the literature: reported case studies on specific food items are too many to enumerate here. Andersson¹⁵ offered an early review of methodological issues and peer-reviewed food LCA studies. Fifteen years later, a review by Roy et al.¹⁶ demonstrates the advancements achieved in the field, but also the challenges and issues that have persisted. Additional review papers have highlighted specific food categories including seafood,^{17,18} livestock products,¹⁹ dairy products,²⁰ and fruit.²¹ The International Conference on LCA in the Agri-Food Sector serves as a global forum for the exchange of recent developments in LCA methodology, databases, and tools, as well as applications of LCA to food-production systems and food-consumption patterns. The eighth LCAFood conference took place in October, 2012 in Saint-Malo, France,²² and the 2014 conference is slated to occur in San Francisco, CA.²³

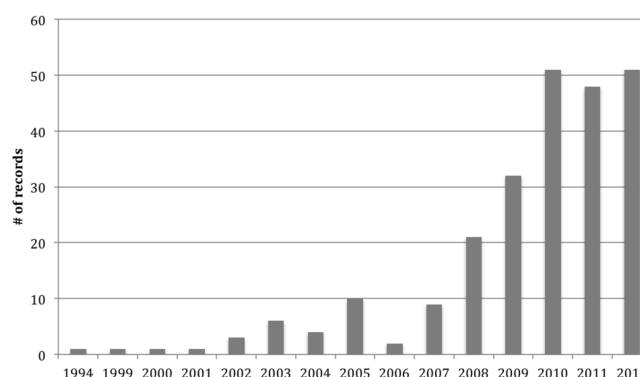


Figure 1. Histogram of the number of articles mentioning “food” and “life cycle assessment”, demonstrating a distinct growth in recent years (search on Web of Science for topic = (food and “life cycle assessment”) resulting in 351 records, and manually culled to 242 records in order to remove inappropriate matches.).

2.1. Production-Oriented Approach. Historically, the point of view in food LCA studies has been to identify opportunities to improve environmental efficiency in food production—decreased environmental impact per unit output (i.e., Garnett’s⁹ efficiency oriented perspective), with a variety of research aims, as shown to the left of the dashed line in Figure 2. Research questions may focus on agricultural production, demonstrating differences in, for example, intensity and type of production methods.²⁴ Extending system boundaries to include processing permits comparisons of food production methods, such as, for example, the scale of bread baking (home baking vs local bakeries vs industrial production).²⁵ Other studies look to highlight environmental hotspots across the entire supply chain, as with, for example, multi-ingredient food products such as tomato catsup.²⁶ LCA has also been used to compare different food items, identifying those foods with the most significant impact: for example, Reijnders and Soret²⁷ compare the environmental impact of different protein sources.

2.2. Defining Functional Unit. LCA is a relative assessment method, and the basis for relative comparison—the functional unit—in food LCAs is a persistent methodological challenge. The functional unit quantifies an identified function of the system under study, and provides the reference to which system inputs and outputs are related. This relative basis allows comparisons of LCA results across alternative systems or scenarios that provide the same function. Succinctly and quantitatively defining the function of a food is challenging, however, and as a result, most LCA studies of food items have utilized the system reference flow (weight or volume) as the functional unit.²⁸ Such mass or volume based functional units are sufficient for many research aims such as identifying system hotspots or evaluating alternative production methods. But when evaluations are made across disparate food types that may constitute different nutritional roles in the diet, an alternative approach to the functional unit is needed. Food serves a variety of functions: it provides pleasure in the form of taste and aesthetics; it plays a role in defining culture and is an avenue for social interaction; it can have emotional and psychological value. For environmental impact studies, and for the purposes of this review, however, it is reasonable to assume that supplying nutrition is the primary function of food consumption; the ideal functional unit basis for diet comparisons should therefore be nutritionally based. The methodological

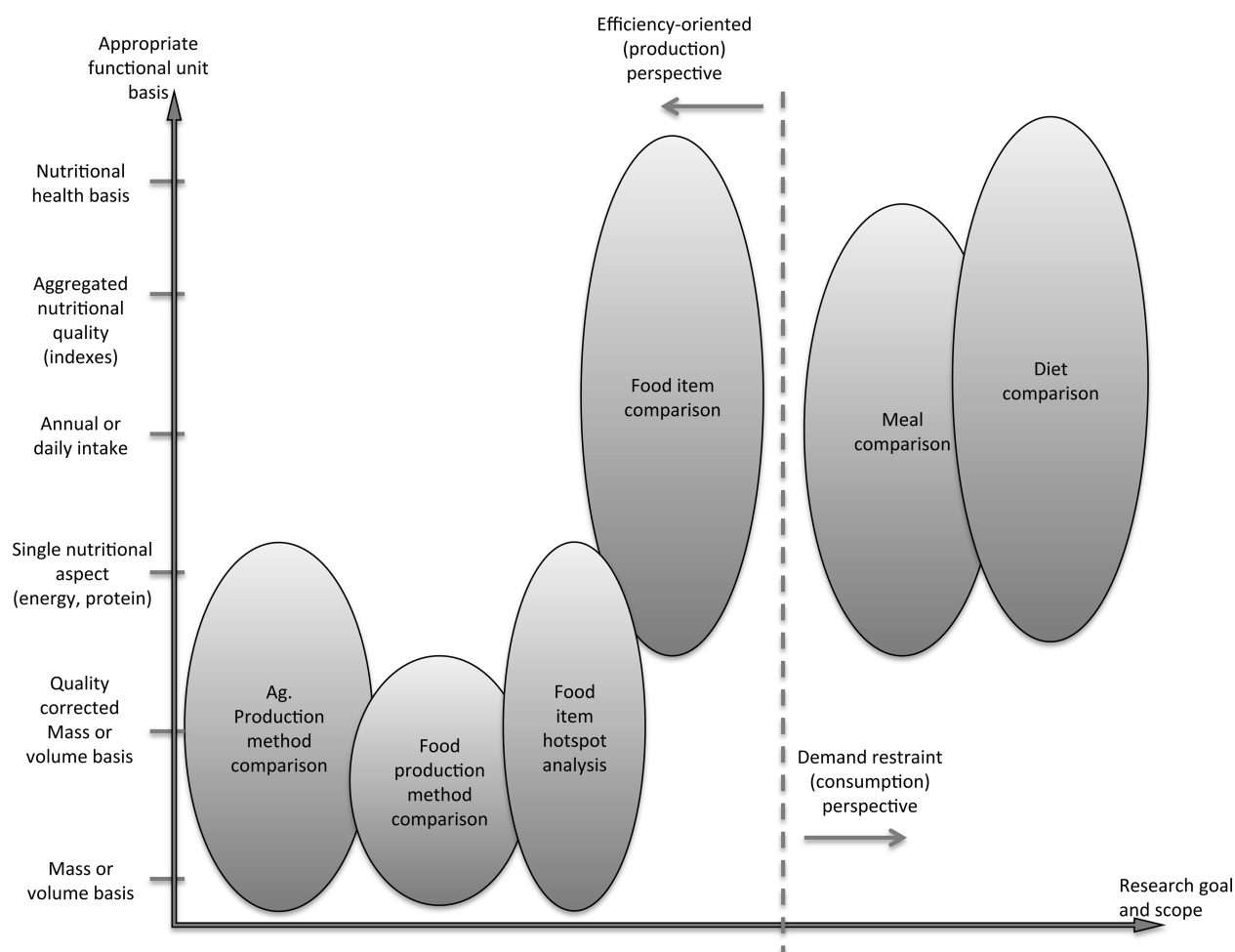


Figure 2. Conceptual map showing the range of appropriate functional unit choices for differing food-related LCA research questions.

challenge of quantitatively linking the nutritional function of foods to their environmental impact in order to inform diet choices forms the basis of this review paper.

Figure 2 offers a conceptual map of reasonable functional unit approaches for different research questions in food related LCAs. The choice of functional unit is dependent on a large number of parameters including the extent of system boundaries (e.g., focused on agricultural production vs including processing, distribution, preparation and consumption), whether it is an internal (single system) or comparative (multiple system) study, the intended stakeholder audience, and of course the particular aim of the study. There is a large body of reported exploration into nutritionally based functional units in the LCAs of food items, but no consistent solution has emerged. Schau and Fet²⁸ review functional unit approaches and recommend a “quality corrected functional unit” that takes the nutrient content of the food products into account. As Schau and Fet point out, this can be relevant for more than comparisons between products, as agricultural production methods can have profound effects on the nutritional quality of foods (e.g., N fertilizer rates affecting the protein content of wheat²⁹). A quality corrected functional unit—“fat and protein corrected milk”—has become the standard in LCAs of fluid milk, accounting for the major nutritional components that are otherwise masked (on a strict weight or volume basis) by their water carrier.³⁰ Functional units based on a single nutritional aspect (e.g., protein content or caloric energy) serve a role for

particular inquiries. For example, various studies comparing food items on the basis of the protein delivered find that plant-based foods have significantly less impact on the environment than animal based foods.^{27,31,32} Nutritional quality is complex and multidimensional, however, and recent interest in identifying foods and diets that are both healthy and environmentally sustainable³³ demands a more nuanced comparative basis. Smedman et al.³⁴ use the nutrient density concept to connect healthy food profiling to GHGE from beverage production. Heller and Keoleian³⁵ and Saarinen³⁶ consider the use of nutritional profiling schemes in making comparisons of the environmental impacts of food items on the basis of their relative contribution to recommended daily nutritional values. Such exploratory efforts can further differentiate foods and affect rankings, but there currently is no standard against which such rankings can be evaluated, making the combined environmental/nutritional indicator difficult to interpret. Table 1 offers a comparative look at the GHGE associated with the production and delivery of various minimally processed food items, along with a nutrient quality indicator, to be discussed in Section 4. Animal-derived foods tend to have greater impact than plant-based foods, with the noted exception of vegetable production in heated greenhouses (hothouse production). Rank comparison across differing functional unit bases offers insight into the importance of the basis for comparison. Imported highly perishable fruits and vegetables requiring airplane transport are another notable

Table 1. Example Life Cycle Greenhouse Gas Emissions (GHGE), Presented on Alternative Functional Unit Bases, and Nutrient Profile Indicator Values for Various Food Items^{a,b}

	per as- sold weight	per serving	per g protein	per kcal food energy	
	kg CO ₂ eq/ kg	kg CO ₂ eq/ serving	kg CO ₂ eq/ 100 g protein	kg CO ₂ eq/ 1000 kcal food energy	weighted nutrient density score
grd. beef	29	2.5	12	13	3 ^c
grd. lamb	26	2.0	10	9.1	−6
cheese	8.6	0.2	3.5	2.1	−62–118 ^d
grd. pork	8.2	0.7	3.2	2.8	0.6
grd. chicken	4.8	0.4	1.8	2.0	19
salmon	3.3	0.2	1.5	2.2	34
egg	3.0	0.2	2.4	2.1	5
tuna	2.6	0.2	1.0	2.2	44
brown rice	1.2	0.05	1.4	0.33	13
white rice	1.2	0.05	1.6	0.33	−3
skim milk	1.1	0.3	3.2	3.2	75 ^e
whole milk	1.1	0.3	3.5	1.8	−7
dry beans	1.0	0.03	0.43	0.30	62
strawberries	0.4	0.03	5.7	1.2	129
broccoli	0.4	0.02	1.3	1.1	164 ^f
orange	0.3	0.04	3.5	0.69	115
tomatoes, field production	0.3	0.04	3.7	1.8	140 ^f
tomatoes, hothouse production	5.3	0.7	61	30	140 ^f
apple	0.3	0.04	11	0.54	66
potato	0.2	0.04	0.81	0.22	18
lettuce	0.2	0.01	2.2	1.4	148 ^f
cabbage	0.1	0.005	0.93	0.49	187 ^f
carrots	0.1	0.003	1.2	0.29	100 ^f
beets	0.1	0.01	0.68	0.19	30
onions	0.1	0.001	0.93	0.23	76 ^f
winter squash	0.09	0.009	1.0	0.24	96
cucumber, field production	0.08	0.005	1.4	0.67	113 ^f
cucumber, hothouse production	1.7	0.3	91	0.45	113 ^f

^aCO₂eq = carbon dioxide equivalents, and includes contributions from all greenhouse gases, expressed relative to the global warming potential of CO₂. Values for a nutrient profile indicator (weighted nutrient density score) are also included to demonstrate food rankings. ^bLife cycle impact data originally compiled by Gonzalez et al.³² Nutritional data from USDA.⁴⁰ Specified and modified as described in Heller and Keoleian.³⁵ Weighted nutrient density scores (WNDS) from Aresenault et al.⁴¹ and from personal communication with Victor Fulgoni. Nutrients included in WNDS are: protein, fiber, calcium, unsaturated fat, Vitamin C, saturated fat, added sugars, and sodium. ^cground beef, 85–89% lean. ^dvalues for cheeses vary widely depending on fat content, from a low of −62 for feta to a high of 118 for fat free mozzarella. ^eCa fortified skim or nonfat milk. ^fvegetables in raw state.

exception to the trends suggested in Table 1. Carlsson-Kanyama et al.³⁷ report the fossil energy needs of supplying fresh strawberries to European markets when transported by plane from the Middle East are four times those grown in Europe; fresh tropical fruits delivered by plane have nine times the energy demand of canned. Similar trends are seen in GHGE of air-freighted fruits and vegetables.^{38,39}

3. FOOD CONSUMPTION-ORIENTED ENVIRONMENTAL IMPACT ASSESSMENT

To address the environmental performance of food from a consumption perspective (right of dashed line in Figure 2), methodological shifts are necessary. First, since foods are rarely consumed in isolation, aggregating food items into real and acceptable meals or diets offers a more realistic view of consumption patterns. Second, it seems imperative to maintain nutritional quality as part of the comparative basis. The defined meal or diet, whether at an individual, household, community or national level, essentially becomes the analytical linkage between environmental impact and nutritional quality. This section reviews demonstrated approaches to evaluating the environmental impact of food consumption patterns (meals or diets) and their key characteristics.

We have identified 48 studies from the English-language literature that consider the environmental impact of food consumption patterns. Those based on an LCA framework are summarized in Table 2. The remaining studies consider diet composition's effect on direct arable land requirements,^{42–46} rely on estimates of energy and emission ratios without a defined LCA framework,^{47–50} utilize the ecological footprint method,^{51–53} employ integrated global environment models^{54,55} or use other methods.^{56,57} While these studies offer valuable insights into the environmental implications of dietary choice, in this review we focus on those studies utilizing LCA methods.

3.1. Methodological Approaches. Table 2 distinguishes between studies that aggregate food consumption at the meal level and diet level. In this paper, meals refer to a collection of foods that might be consumed by an individual in a sitting, whereas diets are meals aggregated or averaged over time and/or over a population. Comparisons of different types of meals (e.g., omnivorous, vegetarian, vegan) can provide illustrative conclusions to the effects of food choices, but are not representative of daily consumption patterns. Diet-level studies can provide insight to the diet choices of whole populations or form the basis for recommendations to consumers. Furthermore, diets tend to consist of a larger number of individual food items, lessening the impact of a particular food choice and providing a stronger basis for connecting food-related environmental impact to nutritional quality.

The majority (81%) of the studies summarized in Table 2 are based on process LCAs⁵⁸ of individual food items, which are then aggregated together into consumption patterns (diets or meals). With this approach, process LCAs of individual foods, as discussed in the previous section, form the building blocks for consumption-oriented studies. At one level, this approach alleviates the functional unit dilemma in LCAs of individual food items, as LCA results presented per kg of food product can easily be aggregated to diets (represented as kg intakes of individual food items). It should be quickly realized, however, that this merely shifts the burden of establishing functional equivalency to the diet level: to make fair comparisons between different diets, a reasonable equalizing basis must be established. The studies labeled with “daily food intake” or “annual food intake” in Table 2 consider diets or meals simply as they are consumed; they do not attempt to establish a nutritionally related (e.g., caloric energy content, protein content) equivalency between diets. Approaches to equalizing diets on a nutritional basis are discussed in Section 5.

Table 2. Summary of Identified Literature Using LCA Approaches to Evaluate the Environmental Impact of Food Consumption

reference	year	geographic scope	LCA type ^a	impact indicators considered ^b	aggregate level ^c	equalizing basis ^d	aim of study
Carlsson-Kanyama ⁸⁹	1998	Sweden	P	GHG, CED	M	E, Pr	hotspot ID, influence of functional unit, ID low GHG foods, is present food consumption “sustainable”?
Kramet ⁸³	1999	Nether-lands	hybrid	GHG	D	AI	GHG of Dutch food consumption based on household expenditure; generic reduction opportunities discussed
Jungbluth ⁶⁶	2000	Switzerland	P	MI (Eco-indicator 95)	D	DI	assist consumers in considering env. aspects of food consumption; focus on consumer POV
Carlsson-Kanyama ³⁷	2003	Sweden	P	CED	M, D	E, AI	present results from experiment w/ more energy efficient diets; compare with current Swedish consumption patterns
Wallén ⁸⁶	2004	Sweden	P	GHG	D	AI	compare GHG of avg Swedish diet and a 1999 defined “sustainable diet”
Sonesson ⁷⁰	2005	Sweden	P	EP, AP, GHG, POCP, CED	M	Same meal	compare meal preparation methods
Baroni ⁸⁰	2007	Italy	P	MI (Eco-Indicator-99 + water impact)	D	E, N	compare omnivorous, vegetarian, vegan diets and conventional vs organic production
Weber ⁶³	2008	US	EIO	GHG	D	AI	compare GHG of food production with food miles; considers avg US household
Davis ⁷¹	2008	Sweden	P	GHG, EP, AP, ODP, CED	M	E	understand environmental impact of integrated food chains and explore improvement measures in postfarm systems
Carlsson-Kanyama ³⁸	2009	Sweden	P	GHG	M	N	review providing overview of GHGs in food production systems, contribution to GHG of food items, and example meals to demonstrate effect of food choices
Munoz ⁶⁷	2010	Spain	P	GHG, AP, EP, CED	D	AI	consider impact of human excretion on food life cycle
Pathak ⁶⁸	2010	India	P	GHG	M, D	DI, N	carbon footprint of Indian food consumption; compare vegetarian and animal based foods
Davis ⁷³	2010	Spain and Sweden	P	renew. E, GHG, POCP, hi NOx areas, ODP, EP, AP	M	E, Pr	explore the benefits of integrating more grain legumes into diet by investigating 4 meals with differing protein sources
Berlin ⁷²	2010	Finland, Norway	P	EP, AP, GHG, CED	M	DI	demonstrate effect of waste decrease at retailer of ready meals
Fazeni ⁸⁴	2011	Austria	P	land use, GHG, CED	D	AI	effects of a shift to recommended diets in Austria; includes agricultural self-sufficiency scenario and energy self-sufficiency for ag
Tukker ⁶⁴	2011	27 countries of EU in 2008	hybrid w/ rebounds	GHG, ODP, AP, human tox., POCP, eco-tox., abiotic resource depletion	D	E	compare impacts between European status quo diets and 3 simulated “healthy” diet baskets
Virtanen ⁶¹	2011	Finland	P, EIO	GHG	M, D	E	ID hotspots, guide consumers in food consumption choices, provide policymakers w/ tool for monitoring impacts on climate change
Berners-Lee ³⁹	2012	UK	P	GHG	D	DI	compare GHG associated with different types of diets
Vieux ⁸⁷	2012	France	P	GHG	D	DI ^e	estimate GHG associated with self-selected diets and evaluate impact of modifying dietary structures
Meier ⁸⁵	2012	Germany	P	GHG, NH3, Land use, water use	D	AI	quantify diet-related environmental impact based on gender differences in typical diets
Macdiarmid ⁸¹	2012	UK	P	GHG	D	E, N	linear programming to answer: can a reduction in GHGs be achieved while meeting dietary requirements for health?
Jungbluth ⁶²	2012	Switzerland	P, EIO	MI (Ecological scarcity)	D		top down (EIO-LCA) and bottom up look at household consumption; evaluate coarse reduction strategies
Kernebeek ⁹⁰	2012	various	P	GHG	D	N	use nutrient quality score “normalization” to evaluate diets reported in literature
Vieux ⁸⁸	2013	France	P	GHG	D	DI ^e	analyze relationship between nutritional quality of self-selected diets and GHGE
Saarién ⁷⁴	2012	Finland	P	GHG, EP	M	DI, N	develop food-related communication tool for upper elementary education
Sanfilippo ⁹¹	2012	Italy	P	CED, GHG, ODP, POCP, AP, EP	M	DI	evaluate environmental impacts of a normal workday; compare different meals with transportation options
Rivera ⁷⁵	2012	UK	P	GHG, AP, EP, POCP, ODP, CED, others	M	DI	compare ready-made and homemade product life cycle for same meal
Oudet ⁶⁹	2012	France	P	GHG	D	AI	compare carbon footprint of conventional and organic food consumption patterns
Kagi ⁹²	2012	Switzer-land	P	MI (ecological scarcity and Eco-Indicator 99)	M	N	explore the use of nutrition indices as functional unit in meal analysis
Saxe ⁷⁹	2013	Denmark	P (consequential)	GHG	D	E, Pr	compare different diets with consequential LCA; consider a number of production and diet scenarios

Table 2. continued

reference	year	geographic scope	LCA type ^a	impact indicators considered ^b	aggregate level ^c	equalizing basis ^d	aim of study
Meier ⁹³	2013	Germany	hybrid	GHG, NH ₃ , land use, blue water use, P use, primary energy use	D	E	compares consumption patterns from 1985 to 1989 with 2006; also compares dietary recommendations and diet styles (ovo-lacto veg., vegan) with 2006 intakes
Wilson ⁸²	2013	New Zealand	P	GHG	M, D		linear programming and scenario approach to optimizing diet nutritional, cost and GHGE profiles

^aP = Process-based LCA; EIO = economic Input-Output based LCA. ^bImpacts: GHG = greenhouse gas emissions; CED = cumulative energy demand; EP = eutrophication potential; AP = acidification potential; ODP = ozone layer depletion potential; POCP = photochemical oxidation potential; MI = multi-impact, single-point indicator. ^cM = meal-level aggregation; D = diet-level aggregation. ^dNormalizing basis: E = caloric food energy; Pr = protein content; AI = annual food intake; DI = daily food intake; N = nutritional equivalence. ^eWhile diets were not normalized, the study does explore the influence of energy content, energy density, and C density of food choices.

An alternative to process-based LCA modeling is the use of economic input output LCA (EIO-LCA). EIO-LCA connects interindustry monetary transactions (economic input-output data) to pollution discharges and nonrenewable resource consumption by industry sectors in order to model the life cycle inventory stage of LCA.⁵⁹ Duchin⁶⁰ offers a conceptual framework for quantitatively analyzing the environmental implications of diet change scenarios through a global EIO model; however, most assessments to date have been performed in a single region construct. Because of the limitations of sector-level resolution, EIO-LCA is often better applied to “top-down” type assessments of national aggregates, as in for example, the carbon footprint of the entire Finnish food chain,⁶¹ or food consumption of the average Swiss⁶² or U.S.⁶³ household. Typical food industry sectors are at the level of “red meat,” “chicken, fish, and eggs,” “dairy products,” “fruits and vegetables,” and “cereals and carbs” and therefore typically do not allow detailed exploration of dietary choices.

A notable example of an EIO-based assessment is the European Commission sponsored study by Tukker et al.^{64,65} which analyzed the environmental impacts of shifts to healthier diets in Europe using a hybrid LCA approach. Hybrid LCA combines EIO-LCA with selective process-LCA data to provide additional resolution in aggregated sectors (e.g., in differentiating between pork and beef in the “meat animals” sector). Utilizing an EIO data set with exceptional sector and product disaggregation (The European Environmentally Extended Input Output Table), Tukker et al. evaluated the food consumption status quo in all 27 EU countries, and compared them with dietary scenarios considered to have positive health impacts. In addition, the study incorporated first-order and second-order rebound effects: scenarios with food costs different than the status quo can cause adjustments in spending in other consumption sectors (first-order rebound), and changed demand in food products may cause price changes, structural changes in primary agricultural sectors, and changes in import and export volumes (second-order rebound). Eight environmental impact categories were normalized and weighted into a single environmental impact score. The study concluded that food consumption under the status quo accounts for 27% of the environmental impacts of total European consumption, and that changes to healthier diets require significant meat and dairy intake reductions in order to influence food-related environmental impacts. Diet changes toward the Mediterranean diet (significantly reduced red meat replaced with chicken, fish, and cereals) reduced the impacts of food consumption by 8%. Second-order rebound modeling suggested that domestic production would likely compensate by increasing exports, however. This implies an even smaller reduction in environmental impact for Europe, and suggests that policies aimed solely at diet change may not be sufficient to achieve impact reduction goals.

3.2. Boundary and Scope of Assessments. Despite the consumption perspective of the studies in Table 2, less than half (40%) include the environmental impacts of the use (consumption) phase within their system boundaries.^{37,38,66–75} The use phase in a food life cycle typically includes storage (refrigeration) and preparation (cooking) of the food; it may also include transportation from a retail outlet to the home. Such use phase practices are very dependent on personal behavior and preferences, and generalizations are challenging. Indeed, many studies that do not include use phase impacts cite data unavailability as the reason. Tukker et al.⁶⁴ justify exclusion

Table 3. Reported Greenhouse Gas Emissions (GHGE) Associated with National Average Per Capita Food Consumption

GHGE (kg CO ₂ eq per capita per day)	reference	notes
7.28	Macdiarmid, 2012 ⁸¹	UK; total food supply emissions distributed across population
4.09 (95% CI: 4.03–4.16)	Vieux, 2013 ⁸⁸	France; based on food intake survey: does not include food waste
7.1	Tukker, 2011 ⁶⁴	EU 27; based on FAO Food Balance Sheets (availability statistics); Hybrid EIO-LCA
8.4	Weber, 2008 ⁶³	U.S.; based on food availability statistics; EIO-LCA
5.6	Saxe, 2013 ⁷⁹	Denmark; based on food availability statistics
7.4	Berners-Lee, 2012 ³⁹	UK; based on self-reported average diet, scaled up to match food energy available (i.e., to include waste)
6.0 (men) 4.2 (women)	Meier, 2012 ⁸⁵	Germany; based on intake survey, adjusted to correspond with food availability statistics. Includes emissions from direct land use change and land use
5.8	Munoz, 2010 ⁶⁷	Spain; based on national food purchases; LCA Includes human excretion and wastewater treatment

of use phase impacts in diet scenario comparisons by suggesting that transport, cooling and preparation of food by final consumers have similar impacts in all diet scenarios; at national aggregate levels, this may be reasonable. A study by Berlin & Sund⁷² considers two different meals and finds the consumer stage contribution (consisting of transport from retailer to household, and electricity use for refrigerator storage and meal heating) contributes 10% to the life cycle GHGE and 13–17% to the life cycle energy consumption. Two separate estimates of the total energy consumption of the U.S. food system place the household-level contribution (household storage and preparation) at 27%⁷⁶ and 32%⁸ of the total.

A number of studies compared meals prepared at home with semi-prepared or ready-to-eat (i.e., industrially prepared) meals,^{70–72,74,75} and in general, found minimal differences in the environmental impact of home-prepared and industrially prepared meals. These studies identify agricultural production as the largest contributor to environmental impact, highlight the importance of food waste at all life cycle stages, and demonstrate that consumer actions can be as important as industrial actions in reducing environmental impact.⁷¹

Munoz et al.⁶⁷ offer a unique contribution by including human excretion (respiration, urine, feces) and subsequent wastewater treatment into the food life cycle. They conclude that the human excretion stage has a minimal contribution to GHGE (after balancing the carbon fixation in photosynthesis of food) and primary energy use, but is an important life cycle stage for eutrophication impacts, accounting for 17% of the eutrophication potential of feeding an average Spaniard for one year. The study also suggests that the consumption phase (storage and cooking at home) contributions to the overall life cycle are (roughly): 14% of GHGE, 23% of primary energy use, 12% of acidification potential, and negligible contribution to eutrophication potential.

Nearly half of the studies in Table 2 consider only the impacts of GHGE, occasionally as a proxy for “environmental impact”. Inevitably, the studies recognize the limitations of this narrow scope and acknowledge that additional impact categories need to be included to draw conclusions on the long-term environmental sustainability of diets. The limited scope of impact categories likely reflects (a) the recent emphasis on GHG reductions within the research community, and (b) inventory data and impact analysis limitations of other impact categories. While climate change is a global impact for which the geospatial source of emissions is unimportant, other impact categories, such as eutrophication, water and land use, and human- and eco-toxicity can have regional and local

impacts, making data sets less applicable across geographies and adding uncertainty to regionally generic impact analysis.^{77,78}

3.3. Defining Diet. The method for constructing meals or diets is a crucial aspect in consumption-oriented studies. Studies have been based on stereotyped meals^{37,61,68} or diets,⁷⁹ diets constructed theoretically to meet nutritional goals and represent a particular value (e.g., “healthful”),^{73,80–82} or diets representative of national or regional averages, often based on national food availability statistics (i.e., (production + imports – exports)).^{39,64,79,80,83–85} A variety of comparisons have been made between current national average diets and “sustainable” or “healthy” diet constructs, attempting to answer whether these alternative diets do indeed represent reduced environmental impact.^{39,64,79,86} While the results of those comparisons vary, the near unanimous conclusion from the studies in Table 2 is that diets based on, or with greater quantities of, animal-based foods, especially red meat and dairy foods, have greater environmental impact. Table 3 offers a comparison of several of the reported GHGE attributable to national average diets. The intention in this comparison is not to suggest real differences in the GHGE intensity of various national diets (although cultural preferences certainly play a role) but to highlight the magnitude and range of reported values. A noted difference in the studies in Table 3 is the basis for constructing the “national average” diet; use of food availability data will generally lead to higher values than use of reported food intakes by individuals.

Vieux et al.^{87,88} offer original contributions as the only known studies to calculate impacts for all self-selected diets of a statistically significant population (diet recall data). Such an approach permits observation of a wide and spontaneous variety of realistic food choices, and allows for a distribution of diet-related impacts across a population, rather than those associated with a single “average” diet. This presents an additional challenge given the large number of food items identified in such diet surveys; the French Individual and National Survey on Food Consumption used by Vieux et al. identified 1314 foods and beverages. To make impact assessment manageable, Vieux et al. selected 73⁸⁷ and 391⁸⁸ of the most widely consumed food items to be representative in 36 food categories, and devised estimation algorithms to correct for the under-coverage of total food intake.

4. NUTRITIONAL QUALITY INDICATORS

A comprehensive means of evaluating nutritional quality is desirable for quantitatively linking environmental impact of dietary patterns to their function of providing nutrition. It is reasonable to assume that “quality” nutrition should contribute to health and wellbeing, but quantitatively measuring the

association between nutrition and health and identifying specific cause-effect factors is extremely complex and challenging. Traditional analyses in nutritional epidemiology have tended to focus on the effect of single foods or nutrients on the risk of developing particular chronic diseases. In the past two decades, limitations in this reductionist approach have led to complementary studies looking at the effect of the overall diet and dietary patterns.^{10,11} Here we summarize current approaches to holistically evaluating nutritional quality at a diet level.

4.1. Diet Quality Indices. A large number of diet quality indices have emerged in recent years, enabling examination of the associations between whole foods, dietary patterns and health. Such diet quality scoring methods have undergone extensive review elsewhere.^{10,94–97} Scoring systems are typically based on current (pre-existing) nutrition guidelines or recommendations (as in the Healthy Eating Index, HEI, which quantifies adherence to the U.S. Dietary Guidelines), but may also derive from a specific dietary pattern that is considered healthy (as in the Mediterranean Diet Score).¹⁰ As such, these scores quantify adherence to the base guidelines or dietary patterns, but have demonstrated variable associations with health outcomes.⁹⁵ For example, higher scores of the HEI-2005 correlated with a 16% lower risk of major chronic disease in large prospective cohort studies, attributable to a 23% lower risk of coronary heart disease and 18% lower risk of diabetes.⁹⁸ (HEI has been updated for conformance to the 2010 Dietary Guidelines for Americans (HEI-2010),⁹⁹ but has not yet been evaluated against health outcomes. The index components of HEI-2005 include total fruit, whole fruit, total vegetables, dark green and orange vegetables and legumes, total grains, whole grains, milk, meat and beans, oils, and moderated intake of saturated fat, sodium and calories from solid fats, alcoholic beverages, and added sugars).⁹⁸ The Alternative Healthy Eating Index (AHEI-2010) has been proposed based on foods and nutrients predictive of chronic disease risk (rather than adherence to dietary guidelines); in the same cohort study, higher AHEI-2010 scores were associated with a 19% lower risk of chronic disease, a 31% lower risk of coronary heart disease, and a 33% lower risk of diabetes.⁹⁸

4.2. Nutrient Profiling. Nutrient profiling is an effort to rank or classify foods based on nutrient composition.¹⁰⁰ Nutrient profiling has potential application in consumer education and dietary guidance, nutrition labeling, regulation of health claims, and evaluation of nutritional quality of food products. Reviews of the various nutrient profiling schemes exist elsewhere;^{100–102} the typical objective is to build a quantitative scoring scheme, aggregating nutritional criteria into a composite index that will accurately characterize each food according to its contribution to the overall balance of the diet. Two more recent developments include the Nutrient Rich Foods Index (NRF)¹⁰³ and the Overall Nutritional Quality Index (ONQI).¹⁰⁴ Composite NRF scores are the arithmetic mean of the percent of recommended daily intakes of nutrients to encourage minus the arithmetic mean of the percent of maximum daily value for nutrients to limit; nutrient values can be based on 100 kcal of food or a reference serving size. NRF9.3, the index variant that includes nine nutrients to encourage (protein, fiber, Vit. A, Vit. C, Vit. E, Ca, Fe, Mg, K) and three nutrients to limit (saturated fat, added sugar, Na), demonstrated the best correlation to the HEI-2005 when evaluated for diets from U.S. National Health and Nutrition Examination Survey (NHANES) populations.¹⁰³ Arsenaault, et

al. demonstrated nonequal weighting of nutrients in an NRF-style profiling scheme by deriving weighting factors from linear regression analysis of nutrient intakes on HEI-2005 using dietary intake data from NHANES 2005–2008.⁴¹ An 8-nutrient Weighted Nutrient Density Score (WNDS) (positive weighting factors for protein, unsaturated fat, fiber, Vit. C, and Ca, and negative weighting factors for Na, saturated fats, and added sugars) explained 65% of the variance in HEI-2005 scores, an improvement over previous nutrient scoring algorithms.⁴¹ Numerical values for WNDS are shown in Table 1 for a variety of foods as an example of food rankings by nutrient profiling. A limitation of this approach is that HEI-2005 itself is a questionable criterion as it is only modestly predictive of major health outcomes.⁹⁸

The ONQI algorithm, implemented commercially in the marketplace as NuVal, incorporates over 30 micro- and macro-nutrient food properties, weighted on the basis of the effects (both promotional and detrimental) of nutrients on health.¹⁰⁴ In general, nutrients are given a trajectory score, which is the ratio of the nutrient density (kcal basis) in the food to the recommended daily intake. Nutrients with favorable effects on health are placed in the numerator of the ONQI algorithm, while those with unfavorable effects are placed in the denominator. Weighting coefficients for each nutrient are based on the prevalence, severity, and strength of association of the nutrient with risk of chronic disease, and were determined through literature review and expert panel consensus;¹⁰⁴ due to NuVal's commercial use, these weighting coefficients remain proprietary. An independent study found that diets scored by the ONQI values of the constituent foods were associated with lower risk of chronic disease and total mortality within two large cohort studies of health professionals over 20 years.¹⁰⁵

While most nutrient profiles have been developed to examine and compare individual food items, examples exist of scoring schemes aggregated to the meal or diet level to provide weighted average food quality scores of dietary patterns,^{103,105} and therefore may prove useful as a diet quality scoring scheme in connection with environmental impact assessments. Nutrient profiles that capture both favorable (nutrients to encourage) and unfavorable (nutrients to limit) effects, especially those indices that are constructed as a difference of favorable and unfavorable effects, while clearly important for correlation with health outcomes, present a potential conceptual challenge if used as a functional unit in an LCA context. LCA typically captures unfavorable effects (impacts) in the numerator, and relates them to favorable effects (societal functions of the system, expressed as a functional unit) in the denominator. At the very least, indices that result in negative values (such as WNDS shown in Table 1) require scaling and/or normalization if used for functional units in LCA.

4.3. Epidemiologic Studies. Epidemiologic studies provide data directly relating foods, nutrients, dietary patterns, or dietary quality indices to human health as an outcome variable, typically expressed as disease incidence or mortality.¹⁰⁶ For example, various sources of protein, such as red meat, fish, and legumes, have been directly compared using incidence of coronary heart disease¹⁰⁷ and diabetes¹⁰⁸ as outcomes. In some studies, a global indicator of adverse health outcomes has been used, such as major chronic disease, which might include incidence of cancer, cardiovascular disease, or death.^{98,105} Limitations of epidemiologic studies include the imperfect nature of diet measurement in a population and the difficulty in ruling out residual confounding, despite statistical adjustments

for multiple variables. In theory, randomized trials can provide more reliable results because confounding is controlled by randomization. However, the evaluation of specific foods or dietary patterns in trials with health outcomes is usually difficult because of poor adherence to assigned diets in studies that may last for many years.¹⁰⁹ An exception is the recent randomized trial demonstrating a reduction in cardiovascular disease by a Mediterranean diet high in monounsaturated fat.¹¹⁰

A common alternative to epidemiologic studies for evaluating the healthfulness of diets is to use the extent to which these diets meet requirements for essential nutrients, for example the percent of Recommended Daily Allowances (RDAs). As described above, consistency with RDAs has sometimes been used to develop or validate dietary quality indices. In principle, these approaches should lead to similar conclusions, but in reality the RDAs are often based on narrow criteria, and may not represent the full impact of whole foods or diets on health. For example, high milk consumption has often been recommended because of its calcium or vitamin D content, but milk has many other constituents that could influence health either positively or negatively. Epidemiologic studies of milk intake in relation to health outcomes provide a summary of all of these constituents and their interactions.

Results from epidemiologic studies have been synthesized within the Global Burden of Disease (GBD) framework¹¹¹ to provide risk factors for underlying causes of health outcomes, including dietary risk factors. The 14 dietary risk factors included in GBD 2010 are diets low in fruits, vegetables, whole grains, nuts and seeds, milk, fish/seafood, fiber, calcium, and polyunsaturated fatty acids, and diets high in red meat, processed meat, sugar-sweetened beverages, trans fatty acids, and sodium.¹¹² Aggregated together, these dietary factors represent the largest contribution to the U.S. burden of disease, even more important than either tobacco smoking, high blood pressure, or high body mass index.¹¹³ GBD risk factors, which are increasingly becoming available on a regional and national level, may offer a health assessment baseline for integration with environmental impact assessment of changes in diets. As a note of caution, the process of developing the global burden of disease and the estimates of underlying causes revealed many gaps in data that highlight the need for better information on diets and health globally.^{112,113}

5. COMBINING NUTRITIONAL QUALITY AND ENVIRONMENTAL IMPACT ASSESSMENT

Efforts to date to account for nutrition in food consumption-oriented LCAs fall into two general categories: designing or modifying comparative meals or diets such that they provide equivalent levels of relevant nutrients; or employing nutritional quality indices (either as functional unit or through coupled evaluation with environmental impacts) as indicators of supplied nutrition.

A number of the papers identified in Table 2 use energy content, protein content, or both as a standardization strategy. For example, Davis et al.⁷³ compare meals that differ in the choice of protein source. Each meal is constructed such that they provide the same (or similar) amounts of protein, energy, and fat, and that the overall size and proportion between meal components are reasonable. Such an approach in essence normalizes the macro-nutrient aspects to allow a fairer comparison of environmental impact. Saxe et al.⁷⁹ take a similar approach in comparing the GHGE of three annual diets (the average Danish diet, consisting of over 300 food and

beverage items supplied to the average Dane; a modification of the average diet based on nutritional recommendations; and an alternative diet inspired by preindustrial Nordic diets designed to be healthy, palatable, and environmentally friendly). The diets were evaluated for energy and protein content, and appropriate quantities of cheese, eggs, and apple juice were added so that the diets supplied equivalent levels of energy and protein.

Meier and Christen⁹³ compare the environmental impact of the average diets in Germany in 1985–1989 and 2006 with two food-based dietary recommendations and two dietary styles (ovo-lacto-vegetarian and vegan); all diets were equalized to 2000 kcal person⁻¹ day⁻¹. Environmental data were based on a hybrid EIO-LCA model, system boundaries were set cradle-to-store (i.e., emissions due to food buying, cooking, storage, and waste disposal were not included), and emissions from direct land use change and land use were included. The study carefully differentiates between mean food intake, taken from National Nutrition Survey data, and food supply, taken from German agricultural statistics; food loss and wastage were treated consistently across dietary scenarios by defining food/food-group-specific conversion factors as the ratio of national mean intake to supply. Results from the study show reduced environmental impacts from the average German diet between 1985 and 1989 and 2006 across all considered indicators (GHG, NH₃ emissions, land use, P use, primary energy use) except blue water use. Reductions were driven by shifts in diet, but partly counterbalanced by increased food wastages. The exception of blue water use was attributed to increased intake of fruits from the late 80s to 2006. Relative to 2006 intakes, dietary shifts to diet recommendations, vegetarian or vegan diets show significant impact reductions in all categories except blue water use. Decreases are primarily attributable to a shift away from animal-based foods; increases in blue water use are attributable to increased consumption of seeds and nuts.

Macdiarmid et al.⁸¹ employ linear programming models to identify diets that meet the UK dietary requirements for adult women while minimizing GHGE. Multiple nutritional constraints were applied in their model: constraints with a lower limit (protein, fiber, complex carbohydrates, vitamins, minerals), constraints with an upper limit (sodium, total fat, saturated fatty acids, and non-milk extrinsic sugars) and an equality constraint (energy). Macdiarmid et al. found that “acceptability constraints” were also needed to arrive at diets with a diversity of food types and realistic quantities. Without such constraints, the model arrived at a diet with a 90% reduction in GHGEs against a 1990 UK baseline, but included only seven food items at unrealistic quantities (dominated by large amounts of fortified whole grain breakfast cereal). Applying “acceptability constraints” generated a realistic diet with 52 foods, but only a 36% reduction in GHGE against the baseline.

An alternative approach to incorporating nutritional evaluation into food-related environmental impact assessment involves utilizing various nutrition profiles or nutritional quality indices. Kagi et al.⁹² compared multi-indicator environmental impact scores of real restaurant meals, adjusted by nutritional quality indicators (nutrient density score (NDS) and nutrient rich food index (NRF9.3)) as the functional unit. With simple meals composed of a protein source (beef, poultry, or mushrooms), potatoes and green beans, Kagi et al. found that, while the meat-containing meals always had greater impact, a NDS functional unit basis lessened the differences between meals by increasing the contributions from vegetables

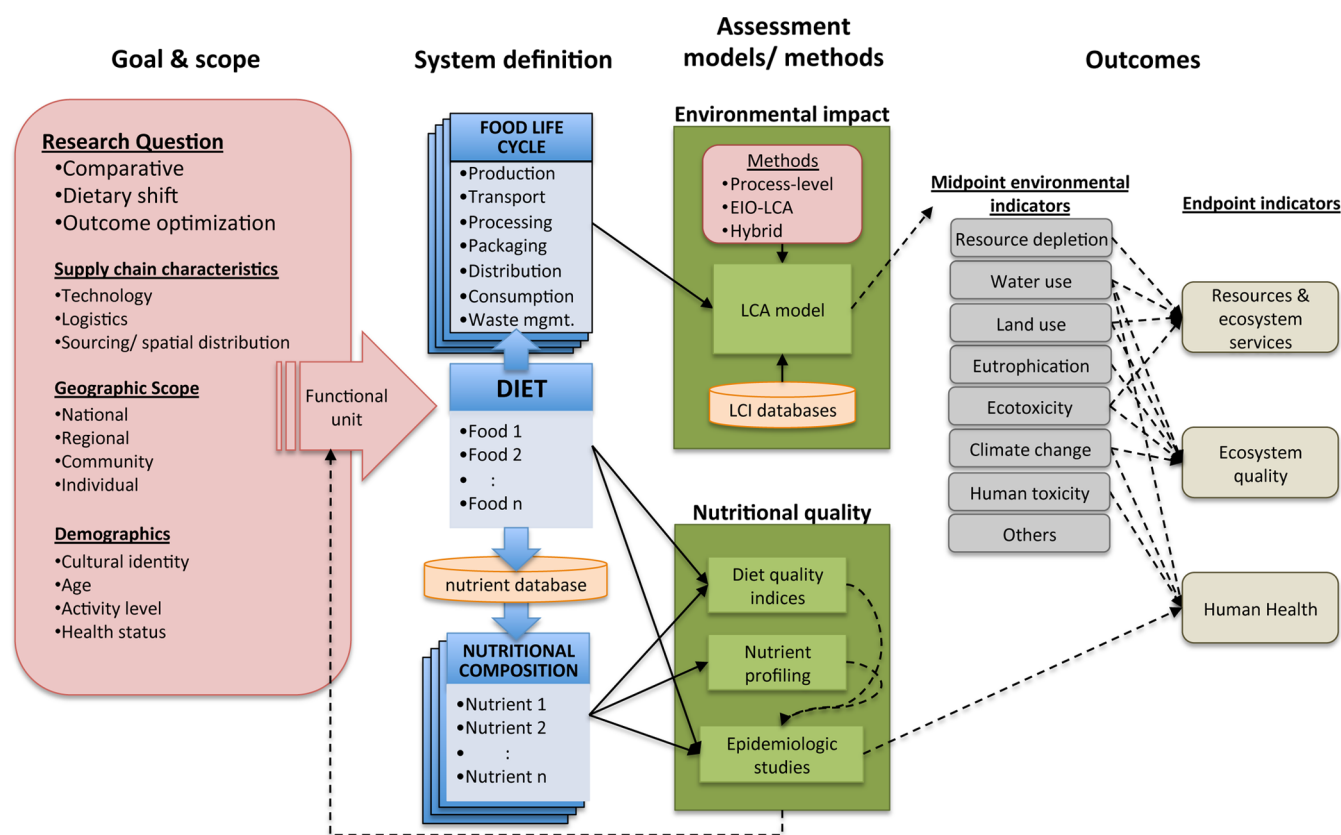


Figure 3. Conceptual framework for diet-level integration of environmental impact and nutritional quality assessments.

(given their lower nutrient density). On the other hand, NRF9.3 as a functional unit emphasized the differences between meals by accentuating the impact of beef (punished for saturated fats).

Kernebeek et al.⁹⁰ apply a diet-level nutrient profiling scheme as a functional unit in comparing the GHGE of diets from published studies. They make an argument for capping nutrient levels at 100% of the recommended daily value (RDV) when using nutrient-based functional units in order to avoid “crediting” overconsumption. For example, if one considers the GHGE related to diets on a protein intake basis, an overconsumption of protein (above the RDV) would lead to decreasing GHGE scores, which seems counter-informative. The same holds true for all “nutrients to encourage” in NRF-style nutrition profiling schemes. The authors concluded that accounting for overall nutritional quality by utilizing a nutrient profile functional unit gives a stronger contrast in GHGE between diets that vary in their amount of animal-based foods.⁹⁰

Vieux et al.⁸⁸ used three indicators of nutritional quality in order to provide a nutritional context to GHGE estimates of self-selected diets of French citizens. The Mean Adequacy Ratio (MAR) was defined as the mean daily percentage of recommended intakes for 20 essential nutrients; the Mean Excess Ratio (MER) as the mean daily percentage of maximum recommended values for three nutrients that should be limited; and energy density (ED) as the ratio between total energy intake and the weight of food intake (excluding beverages). Indicator values for individual diets were compared with sex-specific medians across the entire population. A “high nutritional quality” was defined as one where MAR is above the median, MER is below the median, and ED is below the

median. Four classes of nutritional quality were then established (diets meeting, 3, 2, 1, or 0 of the above properties); these nutritional quality classes were then correlated with the GHGEs associated with particular diets. The researchers concluded that, based on food intake data from a representative sample of French adults, more healthy diets—defined by the nutritional quality classes described—were associated with slightly (but statistically significant) higher GHGEs when diets were normalized by energy intake. This is in spite of the fact that higher nutritional quality as defined by the paper correlated with a higher fraction of dietary energy from plant-based foods; instead, the low GHGE associated with starches and sugar, which likewise scored low on the authors’ measure of nutrient quality, appears to drive the trend. The analysis was limited, however, by lack of actual health outcomes, the fact that nutrients alone do not fully represent the healthfulness of a food, and the dubious importance of energy density as an indicator of health (many foods high in energy density such as nuts are very healthy).

6. DISCUSSION

Diet choices have far reaching implications for both the environmental impact of food systems as well as the health and wellbeing of the consumer. Ample evidence has amassed to provide broad categorical trends regarding environmental impact: agricultural production tends to be the food life cycle stage with the greatest impact (although in totality, household refrigeration and food wastage are significant contributors to food system impacts⁸); animal-based foods tend to have significantly greater impact across most relevant impact categories than do plant based foods (although the type of animals consumed has major influence); out-of-season fruit and

vegetable production requiring heated greenhouses or those products demanding air transport tend to carry significantly greater energy resource demands and carbon footprints. Beyond these gross generalizations, differences become more subtle, can be dependent on production methods and regional conditions, and often may involve trade-offs between environmental impact categories. As individual foods are aggregated to form nutritionally sufficient dietary patterns, still further trade-off dimensions arise (for example, minimizing animal-based foods may be environmentally desirable, but can make meeting certain nutritional requirements more difficult). Add to this the complexity of defining and assuring nutritional quality, as well as social dimensions such as food security and access, community resilience, and animal welfare (not addressed here), and the quest for a healthy food system truly does become a “wicked problem”.¹¹⁴ While there is a common suggestion that dietary changes that are environmentally sustainable also tend to be healthy, this is not necessarily always so. As noted above, Vieux et al.⁸⁸ found that, averaged across the population, French diets that scored higher in the author-defined nutritional quality class also had higher GHGE. Macdiarmid³³ points to anecdotal examples of potential conflicts between health and the environment, including fish intake (recognized as good for health, but can fish stocks support recommended consumption levels?) and low fat dairy and lean meats (recommended for health, but fat and other cuts need to be utilized to avoid wastage). These raise issues that deserve further examination, including the role of sustainable aquaculture in meeting fish demand, and evidence suggesting that adverse health effects of high intakes of dairy and red meat are not necessarily due to their fat content.^{107,108} There is a clear need for databases and methods of analysis that provide linkages between nutritional quality and environmental impact information of food choices to help elucidate such trade-offs.

6.1. Integrating Nutrition. Efforts to date to quantify the environmental impact of diet point toward a need to integrate measures of nutritional quality. LCA remains a popular framework for assessment, but is dependent on an appropriate definition of a functional unit for comparative assessments. Integration with evolving nutrition science methods aimed at capturing nutritional quality appears to hold promise, but will require ongoing dialogue and collaboration between the LCA and nutrition science communities. Such integration can be conceptualized as creating a broader assessment of health. In a general sense, nutrition science aims to understand the link between diet and personal (or public) health. Diet choices indirectly influence public health, as well as ecosystem health and the “health” of natural resource supplies, via environmental interventions brought on by the food supply chain. The conceptual integration of food, sustainability and health has been central to the nutrition ecology¹¹⁵ and environmental nutrition^{116–119} discourse. LCA offers one possible framework, conceptually illustrated in Figure 3, for comprehensively connecting consumption patterns to production implications and quantitatively integrating environmental impact and nutritional health assessments. In the conceptual model in Figure 3, a diet is defined by the research question of interest as well as a host of scoping parameters describing the population and/or geographic boundary in question. Such a diet could represent a national average, intake distributions for a population, nutritional recommendations, or a stereotyped consumption pattern (e.g., the Mediterranean diet, or a vegan diet). Diet combined with supply chain characteristics define

the food life cycles, which are in turn modeled with LCA. Diet and its derivative nutritional composition inform the chosen nutritional quality assessment. It may be desirable to incorporate nutritional quality in the functional unit, or express it as a parallel indicator to environmental effects. Ideally, the nutritional effects on health can be evaluated in a comparable unit to environmental impacts on human health, offering a further degree of integration.

6.2. Addressing Challenges and Needs. Numerous challenges must be addressed in advancing an integrated approach to diet-level assessments. Life cycle assessment is a relatively young methodology, and as such is still undergoing significant development. Many of the developmental needs in LCAs of foods are needs of LCA in general. These include methods for assessment of impacts on ecosystem services from land use and water use,¹²⁰ improvements in uncertainty analysis, and developments of regionalized databases.^{121,122} The acknowledged limitations of a current focus on GHGE among diet-level LCAs must be overcome through further database development and more sophisticated impact assessment methods. Life-cycle sustainability assessment (LCSA) offers a framework for promising future developments by broadening the scope of assessment to include additional dimensions of sustainability (social, economic and environmental), broadening the object of analysis to include sector-level and economy-level assessments, and deepening modeled relations and mechanisms to include greater sophistication and incorporate economic and behavioral mechanisms.^{121,123} Such research edges have already been explored in diet-related studies in, for example, examining economic rebound effects due to shifts in diet.⁶⁴

6.2.1. Land Use Change. Great attention and debate has been paid to the inclusion of emissions from direct (supply chain oriented) land use change¹²⁴ as well as indirect (via market forces) land use change¹²⁵ in LCAs of crop production, particularly biofuel crops. For example, direct land use change effects occur when corn used in a specific ethanol plant is grown on land recently converted from forest or grassland; conversion of Brazilian rainforest to agricultural production driven by increased market demand for corn used in U.S. ethanol production may be considered indirect land use change. Land-use change emissions may also be relevant to consumption-oriented, diet-level assessments, and can significantly influence results and uncertainties. Meier and Christen⁸⁵ account for emissions from direct land use change and land use in their assessment of German diets, and found the impact of land-use related GHG emissions to vary from 16% to 30% of total diet-related emissions, depending on the land-use change scenario considered. Land-use change emissions were associated primarily with animal-based foods, presumably due to the production of soybean and other feed crops. While the International Panel on Climate Change (IPCC) has generated guiding documentation on estimating GHG from land-use change,¹²⁶ significant uncertainties and disagreements on implementation methods exist.

6.2.2. Food Waste and Consumption Data Sets. Food waste is a critical issue in consumption-oriented food LCA studies. Globally, nearly $\frac{1}{4}$ of produced food is lost in the food supply chain.¹²⁷ In the U.S., it is estimated that 10% of available food is lost at the retail level and an additional 19% is lost at the consumer level.¹²⁸ Estimates from the UK indicate that 19% of the food and drink brought into UK homes is wasted, and over 60% of that is avoidable.¹²⁹ Wasted food can grossly affect

environmental impact as it carries all of the upstream burdens without contributing to the functional unit of supplying human nutrition. Measuring food wastage is very challenging, however, especially at the consumer level, and LCA studies typically rely on broad estimates. Recognizing differences in “consumption” data sets is critical in properly accounting for food waste. Sources such as FAO’s Food Balance Sheets¹³⁰ and USDA’s Food Availability Data¹³¹ report per capita food *availability* estimates (production – exports + imports – non-nutritional use); that is, consumer-level waste is a part of this consumption data. Availability estimates are appropriate for calculating national food-related environmental impact (as in Tukker et al.⁶⁴) but overestimate the per capita nutrition intake. USDA has incorporated recent updates of consumer-level food losses¹³² to also provide loss-adjusted food availability,¹³¹ thus providing the basis for a better estimate of per capita dietary intakes in the U.S. Alternatively, dietary intake surveys such as the National Health and Nutrition Examination Survey (NHANES)¹³³ can be used to assess the diet of a population, providing a more realistic distribution of dietary intakes across societal demographics.

When associating environmental impact data sets with nutritional data sets, attention to consistent moisture content and cooking/preparation states is also relevant. Often, LCA data of food items are based on the “at-farm-gate” or “at retailer” state, whereas nutritional data are typically offered “as consumed.” Significant weight changes due to dehydration (e.g., cooking of meat or fish) or rehydration (e.g., cooking of rice or pasta) can lead to substantive errors if not properly accounted. Additional inconsistencies with inedible portions and trimmings (bone-in vs boneless meat, fruits, and vegetables) can also lead to errors.

6.2.3. Developing Food LCA Data Sets. Data availability and quality remain primary obstacles in diet-level environmental impact assessment. The food and agriculture products in current commercially available and widely used data sets such as Ecoinvent¹³⁴ are limited in scope and geographical applicability. There is a distinct lack of geo-spatially explicit food production environmental impact data. As mentioned earlier, this increases uncertainties in the assessment of impact categories that inherently have regional and local impacts. It also raises questions about known differences in standard production practices between countries and regions. A number of the studies in Table 2 that account for imported food (e.g., refs 39 and 79) consider the transport of imports, but due to a lack of specific data, do not differentiate between regional production practices. Efforts are underway to develop a World Food LCA Database with expertise and data from the food industry.¹³⁵ The National Agricultural Library of the USDA has created the LCA Digital Commons as an open access LCA database and tool set.¹³⁶ Such efforts need to be supported and expanded in order to provide the building blocks necessary for comprehensive whole diet assessments and the incorporation of environmental sustainability into dietary guidance policies. Given the important role such data can serve in addressing global food sustainability and security efforts, focus must be placed on providing transparent, open access databases so that diverse stakeholders are working with compatible data sets. Other approaches such as Modular Extrapolation of Agricultural LCA^{137,138} and Multiregional input-output tables¹³⁹ may prove useful in expanding regionally explicit assessments of agricultural and food production. Further, there can be notable differences in production and supply chain practices for the

same food. In addition to “field” vs “hothouse” production and land/water vs air-freight, common examples include organic vs conventionally grown vegetables, grass-fed vs grain-fed beef, or farmed vs wild-caught fish. An ideal data set for consumption-oriented comparisons would include the effects of differences in production practices on both environmental impact and nutrition/health.

6.2.4. Valuation. Additional challenges exist in parallel between environmental impacts of the food system life cycle and health impacts of nutrition. Analyses of both outcomes seek to characterize the performance of complex, interconnected systems through a manageable set of indicators. In communicating performance to decision makers, it is often desirable to combine multiple objectives or dimensions into a single composite score that reasonably weights the value of different objectives. Such valuation, while often based in natural, social, and behavioral sciences and economics, inevitably involves subjective value choices, and with it, controversy. In life cycle impact assessment, the valuation is applied as weighting factors reflecting the relative importance of various environmental impact categories; this allows normalized indicators to be aggregated into a single-score environmental impact. Concerns with weighting approaches are discussed in reviews elsewhere.^{78,140,141} With nutritional quality indicators, valuation occurs in choice of a cutoff value for each index item in the indicator, in the scoring system adopted, and in weighting the components to the total score (typically weighted equally, which is in itself a value choice).^{10,95} Formulation of nutrient profiling schemes involves similar valuation choices. For example, weighting of nutritional components in the ONQI algorithm, while based on literature review of the prevalence and severity of health conditions and the association between the nutrient in question and the condition, ultimately are the result of expert judgment and remain proprietary.¹⁰⁴ It is likely that subjective valuation is unavoidable in developing aggregate indicators of the complexity involved in environmental and nutritional health. It is critical, however, that the basis, structure and impact of such choices be validated, scrutinized and fully communicated to all stakeholders. All of the nutritional indicators considered in this review are reliant on sets of dietary variables thought to be informative of health; these often evolve with scientific understanding and may even be influenced by politics. Consequently, we must recognize that functional unit definitions for life cycle assessment of food systems based on these nutritional indicators will evolve and assessments will need to be recalibrated accordingly. Such challenges are superimposed on broader challenges in nutritional epidemiology, such as determining the effects of diet at different periods in the human life span and with different latencies, accounting for the effects of errors in measuring diet, and dealing with the dynamic nature of our food supply.

6.3. Next Steps. This review summarizes the progress to date in evaluating the environmental impact of dietary patterns through a LCA framework. It is widely recognized that diet plays an important role in sustainable consumption, and sound science-based guidance is required as individuals, industries, and policy-makers address burgeoning environmental challenges. Continued effort is needed to develop and expand life cycle inventory data sets to include regional specificity and additional impact categories (water use, land use, eutrophication, human and eco-toxicity). A major challenge in moving forward is establishing an appropriate comprehensive measure of nutritional quality and/or dietary health effects that can be

coupled with LCA for comparative assessments of dietary patterns. Approaches from nutrition science presented here may serve as a starting point, but continued interdisciplinary dialogue and research is encouraged. Further exploration is needed of integrated environmental and nutritional health assessments in the framework presented here. In particular, advances in analytic linkages and approaches to evaluating multi-objective trade-offs should be encouraged. Evaluation of concrete diet(s) within the integrated framework suggested is a crucial next step. While the practical application of the framework must overcome challenges highlighted, the linkages and key modeling parameters identified can serve to help shape future developments of both fields in improving environmental sustainability and nutritional health of food systems. Establishing methodological guidance and standardization aimed specifically at diet-level assessments will improve consistency and comparability. Recent steps in this direction include the ENVIFOOD Protocol, a proposed harmonized framework assessment methodology for the environmental assessment of food and drink products by the European Food Sustainable Consumption and Production Round Table.¹⁴² A deeper understanding of dietary choices through integrated environmental and nutritional assessments offers a basis for better aligning environmental and health objectives of our food system at a variety of policy levels. While sustainable intensification opportunities—producing more with less—certainly exist within today's food system, attention must also be given to the potential that dietary changes can play in addressing health and environmental problems together. Progress toward a “nutrition-driven food system that sits within environmental limits”⁹ will require concerted input from multiple disciplines.

AUTHOR INFORMATION

Corresponding Author

*Email: mcheller@umich.edu.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge contributions from Olivier Jolliet and Victor Fulgoni III in developing the framework presented. Discussions during an Institute of Medicine of the National Academies workshop in April, 2012 titled “Exploring the True Costs of Food” also informed the development of this paper.

REFERENCES

- (1) Hammond, R. A.; Dubé, L. A systems science perspective and transdisciplinary models for food and nutrition security. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109* (31), 12356–12363.
- (2) Godfray, H. C. J.; Beddington, J. R.; Crute, I. R.; Haddad, L.; Lawrence, D.; Muir, J. F.; Toulmin, C. Food security: The challenge of feeding 9 billion people. *Science* **2010**, *327* (5967), 812–818.
- (3) Foley, J. A.; Ramankutty, N.; Brauman, K. A.; Cassidy, E. S.; Gerber, J. S.; Johnston, M.; Zaks, D. P. M. Solutions for a cultivated planet. *Nature* **2011**, *478* (7369), 337–342.
- (4) Garnett, T. Where are the best opportunities for reducing greenhouse gas emissions in the food system (including the food chain)? *Food Policy* **2011**, *36*, S23–S32.
- (5) Jägerskog, A.; Jonch Clausen, T. *Feeding a Thirsty World—Challenges and Opportunities for a Water and Food Secure Future*, Report no. 31; SIWI: Stockholm, 2012.
- (6) Diaz, R. J.; Rosenberg, R. Spreading dead zones and consequences for marine ecosystems. *Science* **2008**, *321* (5891), 926–929.
- (7) Canfield, D. E.; Glazer, A. N.; Falkowski, P. G. The evolution and future of earth's nitrogen cycle. *Science* **2010**, *330* (6001), 192–196.
- (8) Heller, M. C.; Keoleian, G. A. Assessing the sustainability of the U.S. food system: A life cycle perspective. *Agric. Syst.* **2003**, *76* (3), 1007–1041.
- (9) Garnett, T. Food sustainability: Problems, perspectives and solutions. *Proc. Nutr. Soc.* **2013**, *72* (01), 29–39.
- (10) Waijers, P. M.; Feskens, E. J.; Ocké, M. C. A critical review of predefined diet quality scores. *Br. J. Nutr.* **2007**, *97* (02), 219–231.
- (11) Willett, W. *Nutritional Epidemiology*; Oxford University Press, 2012; Vol. 40.
- (12) Curran, M. A. *Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products*; Wiley, 2012.
- (13) *Environmental Management - Life Cycle Assessment - Principles and Framework*, ISO 14040; International Organization for Standardization, 2006.
- (14) *Environmental Management - Life Cycle Assessment - Requirements and Guidelines*, ISO 14044; International Organization for Standardization, 2006.
- (15) Andersson, K.; Ohlsson, T.; Olsson, P. Life cycle assessment (LCA) of food products and production systems. *Trends Food Sci. Technol.* **1994**, *5* (5), 134–138.
- (16) Roy, P.; Nei, D.; Orikasa, T.; Xu, Q. Y.; Okadome, H.; Nakamura, N.; Shiina, T. A review of life cycle assessment (LCA) on some food products. *J. Food Eng.* **2009**, *90* (1), 1–10.
- (17) Henriksson, P. G.; Guinée, J.; Kleijn, R.; Snoo, G. Life cycle assessment of aquaculture systems—A review of methodologies. *Int. J. Life Cycle Assess.* **2012**, *17* (3), 304–313.
- (18) Pelletier, N.; Ayer, N.; Tyedmers, P.; Kruse, S.; Flysjo, A.; Robillard, G.; Sonesson, U. Impact categories for life cycle assessment research of seafood production systems: Review and prospectus. *Int. J. Life Cycle Assess.* **2007**, *12* (6), 414–421.
- (19) de Vries, M.; de Boer, I. J. M. Comparing environmental impacts for livestock products: A review of life cycle assessments. *Livestock Sci.* **2010**, *128* (1–3), 1–11.
- (20) Milani, F. X.; Nutter, D.; Thoma, G. Invited review: Environmental impacts of dairy processing and products: A review. *J. Dairy Sci.* **2011**, *94* (9), 4243–4254.
- (21) Cerutti, A. K.; Bruun, S.; Beccaro, G. L.; Bounous, G. A review of studies applying environmental impact assessment methods on fruit production systems. *J. Environ. Manage.* **2011**, *92* (10), 2277–2286.
- (22) *8th International Conference on Life Cycle Assessment in the Agri-Food Sector: Proceedings*; INRA: Saint-Malo, France, 2012; <https://colloque4.inra.fr/lcafood2012/Proceedings>.
- (23) American Center for Life Cycle Assessment, LCA of Foods 2014. <http://www.lcacenter.org/lca-of-foods-2014.aspx>.
- (24) Haas, G.; Wetterich, F.; Kopke, U. Comparing intensive, extensified and organic grassland farming in southern Germany by process life cycle assessment. *Agric., Ecosyst. Environ.* **2001**, *83* (1–2), 43–53.
- (25) Andersson, K.; Ohlsson, T. Life cycle assessment of bread produced on different scales. *Int. J. Life Cycle Assess.* **1999**, *4* (1), 25–40.
- (26) Andersson, K.; Ohlsson, T.; Olsson, P. Screening life cycle assessment (LCA) of tomato ketchup: A case study. *J. Clean. Prod.* **1998**, *6* (3–4), 277–288.
- (27) Reijnders, L.; Soret, S. Quantification of the environmental impact of different dietary protein choices. *Am. J. Clin. Nutr.* **2003**, *78* (3), 664S–668S.
- (28) Schau, E. M.; Fet, A. M. LCA studies of food products as background for environmental product declarations. *Int. J. Life Cycle Assess.* **2008**, *13* (3), 255–264.
- (29) Charles, R.; Jolliet, O.; Gaillard, G.; Pellet, D. Environmental analysis of intensity level in wheat crop production using life cycle assessment. *Agric., Ecosyst. Environ.* **2006**, *113* (1), 216–225.

- (30) A Common Carbon Footprint Approach for Dairy: The IDF Guide to Standard Lifecycle Assessment Methodology for the Dairy Sector, IDF 445/2010; International Dairy Federation, 2010.
- (31) de Boer, J.; Helms, M.; Aiking, H. Protein consumption and sustainability: Diet diversity in EU-15. *Ecol. Econ.* **2006**, *59* (3), 267–274.
- (32) Gonzalez, A. D.; Frostell, B.; Carlsson-Kanyama, A. Protein efficiency per unit energy and per unit greenhouse gas emissions: Potential contribution of diet choices to climate change mitigation. *Food Policy* **2011**, *36* (5), 562–570.
- (33) Macdiarmid, J. I. Is a healthy diet an environmentally sustainable diet? *Proc. Nutr. Soc.* **2013**, *72* (01), 13–20.
- (34) Smedman, A.; Lindmark-Månsson, H.; Drewnowski, A.; Edman, A. K. M. Nutrient density of beverages in relation to climate impact. *Food Nutr. Res.* **2010**, *54*.
- (35) Heller, M. C.; Keoleian, G. A. A novel nutrition-based functional equivalency metric for comparative life cycle assessment of food, In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, 2012; pp 401–406.
- (36) Saarinen, M. Nutrition in LCA: Are nutrition indexes worth using? In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, 2012; pp 389–394.
- (37) Carlsson-Kanyama, A.; Ekstrom, M. P.; Shanahan, H. Food and life cycle energy inputs: Consequences of diet and ways to increase efficiency. *Ecol. Econ.* **2003**, *44* (2–3), 293–307.
- (38) Carlsson-Kanyama, A.; Gonzalez, A. D. Potential contributions of food consumption patterns to climate change. *Am. J. Clin. Nutr.* **2009**, *89* (5), S1704–S1709.
- (39) Berners-Lee, M.; Hoolohan, C.; Cammack, H.; Hewitt, C. N. The relative greenhouse gas impacts of realistic dietary choices. *Energy Policy* **2012**, *43*, 184–190.
- (40) USDA, What's In The Foods You Eat Search Tool, v. 5.0. <http://www.ars.usda.gov/Services/docs.htm?docid=17032>.
- (41) Arsenault, J. E.; Fulgoni, V. L.; Hersey, J. C.; Muth, M. K. A novel approach to selecting and weighting nutrients for nutrient profiling of foods and diets. *J. Acad. Nutr. Diet.* **2012**, *112* (12), 1968–1975.
- (42) Gerbens-Leenes, P.; Nonhebel, S. Consumption patterns and their effects on land required for food. *Ecol. Econ.* **2002**, *42* (1), 185–199.
- (43) Gerbens-Leenes, P.; Nonhebel, S.; Ivens, W. A method to determine land requirements relating to food consumption patterns. *Agric., Ecosyst. Environ.* **2002**, *90* (1), 47–58.
- (44) Gerbens-Leenes, W.; Nonhebel, S. Food and land use. The influence of consumption patterns on the use of agricultural resources. *Appetite* **2005**, *45* (1), 24–31.
- (45) Peters, C. J.; Wilkins, J. L.; Fick, G. W. Testing a complete-diet model for estimating the land resource requirements of food consumption and agricultural carrying capacity: The New York State example. *Renewable Agric. Food Syst.* **2007**, *22* (2), 145–153.
- (46) Zhen, L.; Cao, S.; Cheng, S.; Xie, G.; Wei, Y.; Liu, X.; Li, F. Arable land requirements based on food consumption patterns: Case study in rural Guyuan District, Western China. *Ecol. Econ.* **2010**, *69* (7), 1443–1453.
- (47) Eshel, G.; Martin, P. A., Diet, energy, and global warming. *Earth Interact.* **2006**, *10*.
- (48) Eshel, G.; Martin, P. A.; Bowen, E. E., Land use and reactive nitrogen discharge: Effects of dietary choices. *Earth Interact.* **2010**, *14*.
- (49) Pimentel, D.; Pimentel, M. Sustainability of meat-based and plant-based diets and the environment. *Am. J. Clin. Nutr.* **2003**, *78* (3), 660S–663S.
- (50) Coley, D. A.; Goodliffe, E.; Macdiarmid, J. The embodied energy of food: The role of diet. *Energy Policy* **1998**, *26* (6), 455–459.
- (51) Chen, D.; GAO, W.; Chen, Y.; Zhang, Q. Ecological footprint analysis of food consumption of rural residents in China in the latest 30 years. *Agric. Food Sci. Procedia* **2010**, *1*, 106–115.
- (52) Collins, A.; Fairchild, R. Sustainable Food consumption at a sub-national level: An ecological footprint, nutritional and economic analysis. *J. Environ. Policy Plann.* **2007**, *9* (1), 5–30.
- (53) Vintila, I. Ecological footprint evaluation of improved menus using fishery products in the daily diets of UGAL students. *J. Environ. Prot. Ecol.* **2011**, *12* (4A), 2268–2274.
- (54) Popp, A.; Lotze-Campen, H.; Bodirsky, B. Food consumption, diet shifts and associated non-CO₂ greenhouse gases from agricultural production. *Global Environ. Change* **2010**, *20* (3), 451–462.
- (55) Stehfest, E.; Bouwman, L.; van Vuuren, D. P.; den Elzen, M. G. J.; Eickhout, B.; Kabat, P. Climate benefits of changing diet. *Clim. Change* **2009**, *95* (1–2), 83–102.
- (56) Risku-Norja, H.; Hietala, R.; Virtanen, H.; Ketomäki, H.; Helenius, J. Localisation of primary food production in Finland: Production potential and environmental impacts of food consumption patterns. *Agric. Food Sci.* **2008**, *17* (2), 127–145.
- (57) Marlow, H. J.; Hayes, W. K.; Soret, S.; Carter, R. L.; Schwab, E. R.; Sabate, J. Diet and the environment: Does what you eat matter? *Am. J. Clin. Nutr.* **2009**, *89* (5), S1699–S1703.
- (58) Suh, S.; Huppes, G. Methods for life cycle inventory of a product. *J. Clean. Prod.* **2005**, *13* (7), 687–697.
- (59) Matthews, H. S.; Small, M. J. Extending the boundaries of life-cycle assessment through environmental economic input-output models. *J. Ind. Ecol.* **2000**, *4* (3), 7–10.
- (60) Duchin, F. Sustainable consumption of food: A framework for analyzing scenarios about changes in diets. *J. Ind. Ecol.* **2005**, *9* (1–2), 99–114.
- (61) Virtanen, Y.; Kurppa, S.; Saarinen, M.; Katajajuuri, J.-M.; Usva, K.; Maenpää, I.; Nissinen, A. Carbon footprint of food approaches from national input-output statistics and a LCA of a food portion. *J. Clean. Prod.* **2011**, *19* (16), 1849–1856.
- (62) Jungbluth, N.; Itten, R.; Schori, S. Environmental impacts of food consumption and its reduction potentials. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, 2012; pp 104–109.
- (63) Weber, C. L.; Matthews, H. S. Food-miles and the relative climate impacts of food choices in the United States. *Environ. Sci. Technol.* **2008**, *42* (10), 3508–3513.
- (64) Tukker, A.; Goldbohm, R. A.; de Koning, A.; Verheijden, M.; Kleijn, R.; Wolf, O.; Rueda-Cantuche, J. M. Environmental impacts of changes to healthier diets in Europe. *Ecol. Econ.* **2011**, *70* (10), 1776–1788.
- (65) Tukker, A.; Bausch-Goldbohm, S.; Verheijden, M.; Koning, A. d.; Kleijn, R.; Wolf, O.; Dominguez, I. P. *Environmental Impacts of Diet Changes in the EU*, JRC 50544; European Commission Joint Research Centre, Institute for Prospective Technological Studies, 2009.
- (66) Jungbluth, N.; Tietje, O.; Scholz, R. Food purchases: Impacts from the consumers' point of view investigated with a modular LCA. *Int. J. Life Cycle Assess.* **2000**, *5* (3), 134–142.
- (67) Munoz, I.; Canals, L. M. I.; Fernandez-Alba, A. R. Life cycle assessment of the average Spanish diet including human excretion. *Int. J. Life Cycle Assess.* **2010**, *15* (8), 794–805.
- (68) Pathak, H.; Jain, N.; Bhatia, A.; Patel, J.; Aggarwal, P. K. Carbon footprints of Indian food items. *Agric., Ecosyst. Environ.* **2010**, *139* (1–2), 66–73.
- (69) Oudet, N.; Lavelle, P.; Thonier, G.; Pousse, M.; Labouze, E. Carbon footprint of organic vs. conventional food consumption in France. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, Saint-Malo, France, 2012; pp 383–388.
- (70) Sonesson, U.; Mattsson, B.; Nybrant, T.; Ohlsson, T. Industrial processing versus home cooking: An environmental comparison between three ways to prepare a meal. *Ambio* **2005**, *34* (4), 414–421.
- (71) Davis, J.; Sonesson, U. Life cycle assessment of integrated food chains—a Swedish case study of two chicken meals. *Int. J. Life Cycle Assess.* **2008**, *13* (7), 574–584.
- (72) Berlin, J.; Sund, V. *Environmental Life Cycle Assessment (LCA) of Ready Meals*, SIK-Report No. 804 2010; The Swedish Institute for Food and Biotechnology AB, 2010.
- (73) Davis, J.; Sonesson, U.; Baumgartner, D. U.; Nemecek, T. Environmental impact of four meals with different protein sources: Case studies in Spain and Sweden. *Food Res. Int.* **2010**, *43* (7), 1874–1884.

- (74) Saarinen, M.; Kurppa, S.; Virtanen, Y.; Usva, K.; Makela, J.; Nissinen, A. Life cycle assessment approach to the impact of home-made, ready-to-eat and school lunches on climate and eutrophication. *J. Clean. Prod.* **2012**, *28*, 177–186.
- (75) Rivera, X. S.; Orias, N. E.; Azapagic, A. Comparison of the life cycle impacts of ready-made and home-made meals. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, 2012.
- (76) Canning, P.; Charles, A.; Huang, S.; Polenske, K. R.; Waters, A. *Energy Use in the US Food System*, ERR-94; U.S. Department of Agriculture, Economic Research Service, 2010.
- (77) Notarnicola, B.; Tassielli, G.; Renzulli, P. A., Modeling the agri-food industry with life cycle assessment. In *Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products*; Curran, M. A., Ed.; Scrivener Publishing LLC: Salem, MA, 2012; pp 159–184.
- (78) Reap, J.; Roman, F.; Duncan, S.; Bras, B. A survey of unresolved problems in life cycle assessment, Part 2: Impact assessment and interpretation. *Int. J. Life Cycle Assess.* **2008**, *13* (5), 374–388.
- (79) Saxe, H.; Larsen, T.; Mogensen, L. The global warming potential of two healthy Nordic diets compared with the average Danish diet. *Clim. Change* **2013**, *116* (2), 249–262.
- (80) Baroni, L.; Cenci, L.; Tettamanti, M.; Berati, M. Evaluating the environmental impact of various dietary patterns combined with different food production systems. *Eur. J. Clin. Nutr.* **2007**, *61* (2), 279–286.
- (81) Macdiarmid, J. I. Sustainable diets for the future: Can we contribute to reducing greenhouse gas emissions by eating a healthy diet? *Am. J. Clin. Nutr.* **2012**, *96* (3), 632–639.
- (82) Wilson, N.; Nghiem, N.; Ni Mhurchu, C.; Eyles, H.; Baker, M. G.; Blakely, T. Foods and dietary patterns that are healthy, low-cost, and environmentally sustainable: A case study of optimization modeling for New Zealand. *PLoS ONE* **2013**, *8* (3), e59648.
- (83) Kramer, K. J.; Moll, H. C.; Nonhebel, S.; Wilting, H. C. Greenhouse gas emissions related to Dutch food consumption. *Energy Policy* **1999**, *27* (4), 203–216.
- (84) Fazeni, K.; Steinhilber, H., Impact of changes in diet on the availability of land, energy demand, and greenhouse gas emissions of agriculture. *Energy Sustainability Soc.* **2011**, *1*, (6).
- (85) Meier, T.; Christen, O. Gender as a factor in an environmental assessment of the consumption of animal and plant-based foods in Germany. *Int. J. Life Cycle Assess.* **2012**, *17* (5), 550–564.
- (86) Wallén, A.; Brandt, N.; Wennersten, R. Does the Swedish consumer's choice of food influence greenhouse gas emissions? *Environ. Sci. Policy* **2004**, *7* (6), 525–535.
- (87) Vieux, F.; Darmon, N.; Touazi, D.; Soler, L. G. Greenhouse gas emissions of self-selected individual diets in France: Changing the diet structure or consuming less? *Ecol. Econ.* **2012**, *75*, 91–101.
- (88) Vieux, F.; Soler, L.-G.; Touazi, D.; Darmon, N. High nutritional quality is not associated with low greenhouse gas emissions in self-selected diets of French adults. *Am. J. Clin. Nutr.* **2013**, *97* (3), 569–583.
- (89) Carlsson-Kanyama, A. Climate change and dietary choices—How can emissions of greenhouse gases from food consumption be reduced? *Food Policy* **1998**, *23* (3–4), 277–293.
- (90) Kernebeek, H. R. J. v.; Oosting, S. J.; Boer, I. J. M. d. Comparing the environmental impact of human diets varying in amount of animal-source food - the impact of accounting for nutritional quality. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, 2012; INRA: pp 395–400.
- (91) Sanfilippo, S.; Raimondi, A.; Ruggeri, B.; Fino, D. Dietary vs. transport: An analysis of environmental burdens pertaining to a typical workday. *Int. J. Consum. Stud.* **2012**, *36* (2), 133–140.
- (92) Kagi, T.; Zschokke, M.; Dinkel, F. Nutrient based functional unit for meals. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, Saint-Malo, France, 2012.
- (93) Meier, T.; Christen, O. Environmental impacts of dietary recommendations and dietary styles: Germany as an example. *Environ. Sci. Technol.* **2013**, *47* (2), 877–888.
- (94) Kant, A. K. Dietary patterns and health outcomes. *J. Am. Diet. Assoc.* **2004**, *104* (4), 615–635.
- (95) Kourlaba, G.; Panagiotakos, D. B. Dietary quality indices and human health: A review. *Maturitas* **2009**, *62* (1), 1–8.
- (96) Wirt, A.; Collins, C. E. Diet quality-what is it and does it matter? *Public Health Nutr.* **2009**, *12* (12), 2473.
- (97) Ocké, M. C. Evaluation of methodologies for assessing the overall diet: Dietary quality scores and dietary pattern analysis. *Proc. Nutr. Soc.* **2013**, 1–9.
- (98) Chiuve, S. E.; Fung, T. T.; Rimm, E. B.; Hu, F. B.; McCullough, M. L.; Wang, M.; Willett, W. C. Alternative dietary indices both strongly predict risk of chronic disease. *J. Nutr.* **2012**, *142* (6), 1009–1018.
- (99) Guenther, P. M.; Casavale, K. O.; Reedy, J.; Kirkpatrick, S. I.; Hiza, H. A. B.; Kuczyński, K. J.; Krebs-Smith, S. M. Update of the Healthy Eating index: HEI-2010. *J. Acad. Nutr. Diet.* **2013**, *113* (4), 569–580.
- (100) Drewnowski, A.; Fulgoni, V. Nutrient profiling of foods: Creating a nutrient-rich food index. *Nutr. Rev.* **2008**, *66* (1), 23–39.
- (101) Azais-Braesco, V.; Goffi, C.; Labouze, E. Nutrient profiling: Comparison and critical analysis of existing systems. *Public Health Nutr.* **2006**, *9* (5), 613–22.
- (102) Garsetti, M.; de Vries, J.; Smith, M.; Amosse, A.; Rolf-Pedersen, N. Nutrient profiling schemes: Overview and comparative analysis. *Eur. J. Nutr.* **2007**, *46*, 15–28.
- (103) Fulgoni, V. L., III; Keast, D. R.; Drewnowski, A. Development and validation of the nutrient-rich foods index: A tool to measure nutritional quality of foods. *J. Nutr.* **2009**, *139* (8), 1549–1554.
- (104) Katz, D. L.; Njike, V. Y.; Faridi, Z.; Rhee, L. Q.; Reeves, R. S.; Jenkins, D. J.; Ayoob, K. T. The stratification of foods on the basis of overall nutritional quality: The overall nutritional quality index. *Am. J. Health Promot.* **2009**, *24* (2), 133–143.
- (105) Chiuve, S. E.; Sampson, L.; Willett, W. C. The Association between a nutritional quality index and risk of chronic disease. *Am. J. Prev. Med.* **2011**, *40* (5), 505–513.
- (106) Willett, W. *Nutritional Epidemiology*; Oxford University Press, 2012.
- (107) Bernstein, A. M.; Sun, Q.; Hu, F. B.; Stampfer, M. J.; Manson, J. E.; Willett, W. C. Major dietary protein sources and risk of coronary heart disease in women. *Circulation* **2010**, *122* (9), 876–883.
- (108) Pan, A.; Sun, Q.; Bernstein, A. M.; Schulze, M. B.; Manson, J. E.; Willett, W. C.; Hu, F. B. Red meat consumption and risk of type 2 diabetes: 3 cohorts of US adults and an updated meta-analysis. *Am. J. Clin. Nutr.* **2011**, *94* (4), 1088–1096.
- (109) Willett, W. C. The WHI joins MRFIT: A revealing look beneath the covers. *Am. J. Clin. Nutr.* **2010**, *91* (4), 829–830.
- (110) Estruch, R.; Ros, E.; Salas-Salvadó, J.; Covas, M.-I.; Corella, D.; Arós, F.; Lapetra, J. Primary prevention of cardiovascular disease with a Mediterranean diet. *N. Engl. J. Med.* **2013**, *368* (14), 1279–1290.
- (111) Murray, C. J. L.; Ezzati, M.; Flaxman, A. D.; Lim, S.; Lozano, R.; Michaud, C.; Lopez, A. D. GBD 2010: Design, definitions, and metrics. *Lancet* **2012**, *380* (9859), 2063–2066.
- (112) Lim, S. S.; Vos, T.; Flaxman, A. D.; Danaei, G.; Shibuya, K.; Adair-Rohani, H.; Ezzati, M. A comparative risk assessment of burden of disease and injury attributable to 67 risk factors and risk factor clusters in 21 regions, 1990–2010: A systematic analysis for the Global Burden of Disease Study 2010. *Lancet* **2012**, *380* (9859), 2224–2260.
- (113) Murray, C. J.; Abraham, J.; Ali, M. K.; Alvarado, M.; Atkinson, C.; Baddour, L. M.; Birbeck, G. The State of US Health, 1990–2010: Burden of Diseases, Injuries, and Risk Factors. *JAMA, J. Am. Med. Assoc.* **2013**, *310* (6), 591–608.
- (114) Hamm, M. W. Principles for framing a healthy food system. *J. Hunger Environ. Nutr.* **2009**, *4* (3–4), 241–250.
- (115) Leitzmann, C. Nutrition ecology: The contribution of vegetarian diets. *Am. J. Clin. Nutr.* **2003**, *78* (3), 657S–659S.
- (116) Gussow, J. D.; Clancy, K. L. Dietary guidelines for sustainability. *J. Nutr. Educ.* **1986**, *18* (1), 1–5.
- (117) Wallinga, D. Today's food system: How healthy is it? *J. Hunger Environ. Nutr.* **2009**, *4* (3–4), 251–281.

- (118) Lang, T. Reshaping the food system for ecological public health. *J. Hunger Environ. Nutr.* **2009**, 4 (3–4), 315–335.
- (119) Hamm, M. W. Linking sustainable agriculture and public health: Opportunities for realizing multiple goals. *J. Hunger Environ. Nutr.* **2008**, 3 (2–3), 169–185.
- (120) Rack, M.; Valdivia, S.; Sonnemann, G. Life cycle impact assessment—Where we are, trends, and next steps: A late report from a UNEP/SETAC Life Cycle Initiative workshop and a few updates from recent developments. *Int. J. Life Cycle Assess.* **2013**, 1–8.
- (121) Guinee, J. B.; Heijungs, R.; Huppes, G.; Zamagni, A.; Masoni, P.; Buonomici, R.; Rydberg, T. Life cycle assessment: Past, present, and future. *Environ. Sci. Technol.* **2011**, 45 (1), 90–96.
- (122) Finnveden, G.; Hauschild, M. Z.; Ekvall, T.; Guinee, J.; Heijungs, R.; Hellweg, S.; Suh, S. Recent developments in life cycle assessment. *J. Environ. Manage.* **2009**, 91 (1), 1–21.
- (123) Zamagni, A.; Guinee, J.; Heijungs, R.; Masoni, P. Life cycle sustainability analysis. In *Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products*; Curran, M. A., Ed.; Scrivener Publishing LLC, 2012; pp 453–474.
- (124) Fargione, J.; Hill, J.; Tilman, D.; Polasky, S.; Hawthorne, P. Land clearing and the biofuel carbon debt. *Science* **2008**, 319 (5867), 1235–1238.
- (125) Searchinger, T.; Heimlich, R.; Houghton, R. A.; Dong, F.; Elobeid, A.; Fabiosa, J.; Yu, T.-H. Use of U.S. croplands for biofuels increases greenhouse gases through emissions from land-use change. *Science* **2008**, 319 (5867), 1238–1240.
- (126) Intergovernmental panel on climate change. *Good Practices Guidance for Land Use, Land-Use Change and Forestry*, 2003. <http://www.ipcc-nggip.iges.or.jp/public/gpglulucf/gpglulucf.html>.
- (127) Kumm, M.; de Moel, H.; Porkka, M.; Siebert, S.; Varis, O.; Ward, P. J. Lost food, wasted resources: Global food supply chain losses and their impacts on freshwater, cropland, and fertiliser use. *Sci. Total Environ.* **2012**, 438 (0), 477–489.
- (128) Buzby, J. C.; Hyman, J. Total and per capita value of food loss in the United States. *Food Policy* **2012**, 37 (5), 561–570.
- (129) Quested, T.; Parry, A. *New Estimates for Household Food and Drink Waste in The UK*; WRAP, 2011. http://www.fcrn.org.uk/sites/default/files/New_estimates_household_food_drink_0.pdf.
- (130) Food and Agriculture Organization of the United Nations. FAOSTAT. <http://faostat3.fao.org/home/index.html> - HOME.
- (131) United States Department of Agriculture Economic Research Service, Food Availability (Per Capita) Data System. [http://www.ers.usda.gov/data-products/food-availability-\(per-capita\)-data-system.aspx](http://www.ers.usda.gov/data-products/food-availability-(per-capita)-data-system.aspx).
- (132) Muth, M. K.; Karns, S. A.; Nielsen, S. J.; Buzby, J. C.; Wells, H. F. *Consumer-Level Food Loss Estimates and Their Use in the ERS Loss-Adjusted Food Availability Data*, TB-1927; United States Department of Agriculture, Economic Research Service, 2011; <http://www.ers.usda.gov/publications/tb-technical-bulletin/tb1927.aspx>.
- (133) Centers for Disease Control and Prevention, National Health and Nutrition Examination Survey (NHANES). <http://www.cdc.gov/nchs/nhanes.htm>.
- (134) Swiss Center for Life Cycle Inventories, Ecoinvent Database version 2.0. <http://www.ecoinvent.org/home/>.
- (135) Peano, L.; Loerincik, Y.; Humbert, S.; Gaillard, G.; Nemecek, T. The LCA world food database project: Towards more accurate LCAs on agricultural and food products. In *8th International Conference on LCA in the Agri-Food Sector*, Saint-Malo, France, INRA, 2012; p 891.
- (136) USDA National Agricultural Library, LCA Digital Commons. <http://www.lcacommons.gov/>.
- (137) Roches, A.; Nemecek, T.; Gaillard, G.; Plassmann, K.; Sim, S.; King, H.; i Canals, L. M. MEXALCA: A modular method for the extrapolation of crop LCA. *Int. J. Life Cycle Assess.* **2010**, 15 (8), 842–854.
- (138) Nemecek, T.; Weiler, K.; Plassmann, K.; Schnetzer, J. *Geographical Extrapolation of Environmental Impact of Crops by the MEXALCA Method: Unilever-ART Project No. CH-2009-0362 “Carbon and Water Data for Bio-based Ingredients”: Final Report of Phase 2: Application of Method and Results*; Agroscope Reckenholz-Tanikon Research Station ART: Zurich, CH, 2011.
- (139) Lenzen, M.; Kanemoto, K.; Moran, D.; Geschke, A. Mapping the structure of the world economy. *Environ. Sci. Technol.* **2012**, 46 (15), 8374–8381.
- (140) Bare, J. Life cycle impact assessment research developments and needs. *Clean Technol. Environ. Policy* **2010**, 12 (4), 341–351.
- (141) Pennington, D. W.; Potting, J.; Finnveden, G.; Lindeijer, E.; Jolliet, O.; Rydberg, T.; Rebitzer, G. Life cycle assessment Part 2: Current impact assessment practice. *Environ. Int.* **2004**, 30 (5), 721–739.
- (142) European Food Sustainable Consumption and Production Round Table. <http://www.food-scp.eu/>.